

|                 |                                                                                                                                                    |    |     |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|
| <b>1 (a)</b>    | acceleration = change in velocity / time (taken) or rate of change of velocity                                                                     | B1 | [1] |
| <b>(b) (i)</b>  | $v = 0 + at$ or $v = at$                                                                                                                           | C1 |     |
|                 | $(a = 36/19 \Rightarrow) 1.9 (1.8947) \text{ ms}^{-2}$                                                                                             | A1 | [2] |
| <b>(ii)</b>     | $s = \frac{1}{2}(u + v)t$ or $s = v^2/2a$ or $s = \frac{1}{2}at^2$                                                                                 |    |     |
|                 | $= \frac{1}{2} \times 36 \times 19$ $= 36^2/(2 \times 1.89)$ $= \frac{1}{2} \times 1.89 \times 19^2$                                               |    |     |
|                 | $= 340 \text{ m (342 m/343 m/341 m)}$                                                                                                              | M1 | [1] |
| <b>(iii) 1.</b> | $(\Delta KE \Rightarrow) \frac{1}{2} \times 95 \times (36)^2$                                                                                      | C1 |     |
|                 | $= 62\,000 (61\,560) \text{ J}$                                                                                                                    | A1 | [2] |
| <b>2.</b>       | $(\Delta PE \Rightarrow) 95 \times 9.81 \times 340 \sin 40^\circ$ or $95 \times 9.81 \times 218.5$                                                 | C1 |     |
|                 | $= 200\,000 \text{ J}$                                                                                                                             | A1 | [2] |
| <b>(iv)</b>     | work done (by frictional force) = $\Delta PE - \Delta KE$<br>or<br>work done = $200\,000 - 62\,000$ (values from <b>1b(iii) 1.</b> and <b>2.</b> ) | C1 |     |
|                 | (frictional force = $138\,000/340 \Rightarrow$ ) $410 (406) \text{ N}$ [420 N if full figures used]                                                | A1 | [2] |
| <b>(v)</b>      | $-ma = mg \sin 20^\circ - f$ or $ma = -mg \sin 20^\circ + f$                                                                                       | C1 |     |
|                 | $-95 \times 3.0 = 95 \times 3.36 - f$                                                                                                              |    |     |
|                 | $f = 600 (604) \text{ N}$                                                                                                                          | A1 | [2] |